



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

STA1023/MAT1063- TUTORIAL 1

1. A man has 3 different suits, 4 different shirts and 5 different pairs of shoes. In how many different ways can this man wear a suit, a shirt and pair of shoes?
2. In a certain country, license plate numbers have 3 uppercase letters followed by 4 digits. How many different license plate numbers can be formed? (Letters and digits may be repeated).
3. Using the digits 1,2,3 and 5, how many digit numbers can be formed if
 - (a) The first digit must be 1 and repetition of the digits is allowed?
 - (b) The first digit must be 1 and repetition of the digits is not allowed?
 - (c) The number must be divisible by 2 and repetition is allowed?
 - (d) The number must be divisible by 2 and repetition is not allowed?
4. Determine the number of distinguishable arrangements for MILLENNIUM.
5. In how many ways cricket-eleven be chosen out of 15 players? If
 - i) A particular player is always chosen,
 - ii) A particular is never chosen.
6. Consider a group of 20 people. If everyone shakes hands with everyone else, how many handshakes take place?
7. How many ways can 7 individuals sit at a round table? What will be the value if two of them refuses to sit together?
8. How many functions defined on n points are possible if each functional value is either 0 or 1?
9. Show that $\binom{n-1}{r} + \binom{n-1}{r-1} = \binom{n}{r}$ [Pascal's Identity]
10. For which x positive integer stands: ${}^{x-1}C_{x+1} \leq 21$

Submit on or before 07th of July 2026 (Tuesday) by 4.00 p.m.